Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

A realistic upper-limb musculoskeletal model for overall muscle activation estimation

Tristan Barjavel (tristan.barjavel@epfl.ch)
École Polytechnique Fédérale de Lausanne
Tommaso Proietti
Sant'Anna School of Advanced Studies
Camilla Pierella
University of Genoa
Silvestro Micera
École Polytechnique Fédérale de Lausanne

Article

Keywords: Biomechanics, Electromyography (EMG), Muscle activations, Musculoskeletal model, Upper-limb

Posted Date: May 18th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-2802018/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

A realistic upper-limb musculoskeletal model for overall muscle activation estimation

Tristan Barjavel[1,*], Tommaso Proietti[2], Camilla Pierella[3,4], and Silvestro Micera[1,2]

[1]Ecole Polytechnique Federale de Lausanne (EPFL), Bertarelli Foundation Chair in Translational Neuroengineering, Center for Neuroprosthetics and School of Engineering, Geneva, 1202, Switzerland
[2]Scuola Superiore Sant'Anna, Biorobotics Institute and Department of Excellence in Robotics & AI, Pisa, 56025, Italy
[3]University of Genoa, Department of Neurosciences , Rehabilitation, Ophthalmology, Genetics, and Maternal and Children's Sciences (DINOGMI) , Genoa, 16126, Italy
[4]University of Genoa, Department Informatics, Bioengineering, Robotics and Systems Engineering (DIBRIS), Genoa, 16126, Italy
[*]Corresponding author: tristan.barjavel@epfl.ch

ABSTRACT

The human upper-limb can perform complex tasks thanks to the ability of the central nervous system (CNS) to control and modulate the activation of more than 40 muscles. A deeper understanding of the strategies implemented by the CNS to perform these movements could help develop more effective rehabilitation protocols. Unfortunately, given the large number of muscles acting on the shoulder and the arm, and their positions with respect to each other, a simultaneous electromyographic (EMG) recording of all the upper-limb's muscles is not feasible in practice. Numerical musculoskeletal models could represent a useful alternative approach to gather this kind of information. Here, we develop a new realistic upper-limb biomechanical model which uses a combination of few recorded EMG data (up to max N=16) and an inverse dynamics model to predict the overall upper-limb muscle activations (N=42). With data from five healthy subjects performing 3D upper-limb movements, the predictions from this EMG-assisted model were compared to: 1) a full set of 16 arm muscles EMG data, 2) a model using none of the these recorded EMG data (standard static-optimization). While predicted signals from the EMG-assisted and static-optimization led to a 3.4% and 2.3% error respectively in the resulting moment of joint when fed into the movement dynamics, the EMG-assisted method presented a more physiological sharing of the muscle activations and variations of errors with respect of the activations from EMG signals for non used muscles in a comparison leave-one-out method. These results demonstrate the ability of the proposed musculoskeletal model to be used as a tool to evaluate healthy motion data, and open up to the application of such a strategy to analyze impaired individuals motion data and to compare inter-subject activation behaviours in the future.

Keywords Biomechanics, Electromyography (EMG), Muscle activations, Musculoskeletal model, Upper-limb

Introduction

Human healthy individuals can easily control their upper-limb movements despite a complex and redundant musculoskeletal system. In 1967, Nikolai Bernstein[1] hypothesized that the central nervous system (CNS) activates groups of muscles as muscle synergies instead of activating them independently, with each synergy specifying a particular balance of activation across a set of muscles. More recently, this hypothesis has been tested and confirmed by other groups[2–4]. Understanding how the CNS manages to perform complex tasks requires a deep comprehension of how the different muscles activate. Because of the often-overlapping physical location and the large number, simultaneous direct measurement of multiple muscles (e.g. the ones in the upper-limb) via electromyography (EMG) is generally not feasible at a large scale, in particular in a clinical environment. However, EMG is still largely considered the gold standard for assessing muscle activity, being a non-intrusive method with fast setup time. It is important to notice that EMG recordings are able to show the activity of the muscles but do not provide their contribution to the general movement, underestimating the muscles passive forces derived from their sole elongation[5]. Musculoskeletal numerical models could represent an interesting alternative approach to directly probing the muscles[6]. Existing models mainly focus on estimating the force produced by the different muscles, often neglecting the dynamics of muscle activation. Forces are usually computed by optimization of physiological cost-functions and constraints, applied to equations of moment equilibrium at each joint. Most of these optimization criteria are based on muscle-related properties, for instance, muscle stress[7], muscle forces[8], and muscle's energy consumed[9], or the so-called min/max criterion[10,11]. In a few studies, EMG signals were incorporated into lower-limb musculoskeletal models[12,13] and upper-limb musculoskeletal models[14–18], resulting

in more accurate estimates of muscle co-contraction estimations.

In this study, we propose an overall fast muscle activation estimator (42 muscles in total) as a comparative and evaluation tool, that improves muscle co-contraction estimations with the use of a reduced number of recorded EMG signals (up to N=16). The estimator works together with a numerical upper-limb musculoskeletal model that uses a combination of approaches: 1) minimization of the sum of squared muscle stresses, setting an evenly distributed load over the muscles; 2) constraining muscle forces to keep a glenoid stability and to keep their values within their physiological boundaries; 3) minimization of the squared differences in the obtained and experimental joint moments. Physiological boundaries are defined by the contraction dynamics with feasible activations as well as available recorded EMGs. The minimization criteria is different from previous models where authors mostly minimized the overall muscle activations for non-recorded muscles while keeping a low predicted error between joint moments and measured activations in forward dynamics for lower limb[12,19]. with CEINMS toolbox on OpenSim[20]. In our previous previous upper-limb model[16], the minimization was done on the forces, using the null space from the underlying matrix of the muscle moment arms; however, using this strategy a solution could not always be met. Another difference with the state of the art is that the provided cost function to minimize is not subject-specific as it was previously proposed[12,19]. This important feature aims at improving the ability of the model to perform subject comparisons with the future goal of analyzing intra-subject activation evolution to optimize and guide subject-specific rehabilitation therapy.

Despite the different minimization criteria, the model we present is also an improved version of our previous upper-limb models[16,21,22] in terms of computational quickness and efficiency regarding inner processes, such as musculotendon parameters estimations, rigid tendon modelling, signal treatments, and the methods to perform inverse dynamics. The model was tested and validated by processing data from five healthy subjects performing 3D reaching movements using ALEx RS[23], an upper-limb exoskeleton. In particular, we used our model to predict the activities of the upper-limb muscles during the reaching movements and compared them with available EMG signals recorded during the experiments. The general overview of the work is shown in Fig. 1.

Methods

Numerical model

In this work we present a model derived from an existing validated shoulder musculoskeletal model[16,21,22] that was initially developed for predicting both the muscle and the joint reaction forces. The numerical musculoskeletal model was developed in MATLAB (MathWorks, USA). It uses anthropometric cadaverous data but allows for scaling parameters to match the dimensions of each specific subject. Tunable parameters are i) the bone inertial properties, ii) the skeletal morphologies, iii) the muscle architectures; all these features could allow for an improved inter-subject, healthy/impaired muscle activation evaluation in the future[21]. However, subject-specific models require complex modeling workflows[24,25] and well-defined methods, especially for musculotendon parameters estimation. For our study, given the nature of the recorded kinematics (all subjects using the same exoskeleton) and the collected information about the subjects (see following section), a fixed scaling was applied to match a generic model to all subjects, so that all subjects would have a common basis for the model outputs evaluations, based on their movement and their EMG recorded[26]. In the original model, 42 muscle bundles are considered, spanning 11 degrees of freedom (DOF) including the sternoclavicular, acromioclavicular, glenohumeral modeled as ball and sockets joint as well as the elbow flexion-extension and pronation-supination between the thorax, clavicle, scapula, humerus ulna and radius according to the ISB recommendations[27]. The muscles are represented with their centroid lines, from their origin to their insertions, and they lie on defined wrapping objects (Fig. 1). With the movement represented by the model's angular rotations at the joint levels, the different musculotendon lengths, their directions, their moment arms[28], and all inertia parameters can be computed.

Experimental set-up and recorded data

To validate the model, we used EMG and kinematic recordings of five healthy subjects (4 females, 1 male, 60 ± 12 years old, 65 ± 11 kg, 167 ± 9 cm) performing reaching tasks while assisted by an upper-limb exoskeleton (Arm Light Exoskeleton Rehab Station, ALEx RS[23,29]). The full experimental set-up and the description of the subjects are available elsewhere[30]. The robotic assessment consisted of performing center-out 3D reaching movements toward 18 target positions, equally distributed on a sphere (19cm radius). Subjects executed these movements 5 times with the right arm, while seated and wearing ALEx RS, and were always returning to a starting position placed in the center of the target sphere. A monitor in front of the subject displayed the end-effector position, the different targets of the sphere, and the specific target to reach. The center of the spherical workspace position was defined to be aligned with the right acromion of the subject and equidistant between the monitor and the acromion (Fig. 2).

To test the model, the targets considered were the 6 points positioned along the 3 main directional axes from the center of the sphere, corresponding to reaching upwards (target 1), downwards (target 5), to the left (target 7) to the right (target 3), frontwards (target 13) and rearwards (target 10). The robot was only used to record the subject's kinematics in transparent mode, that is it

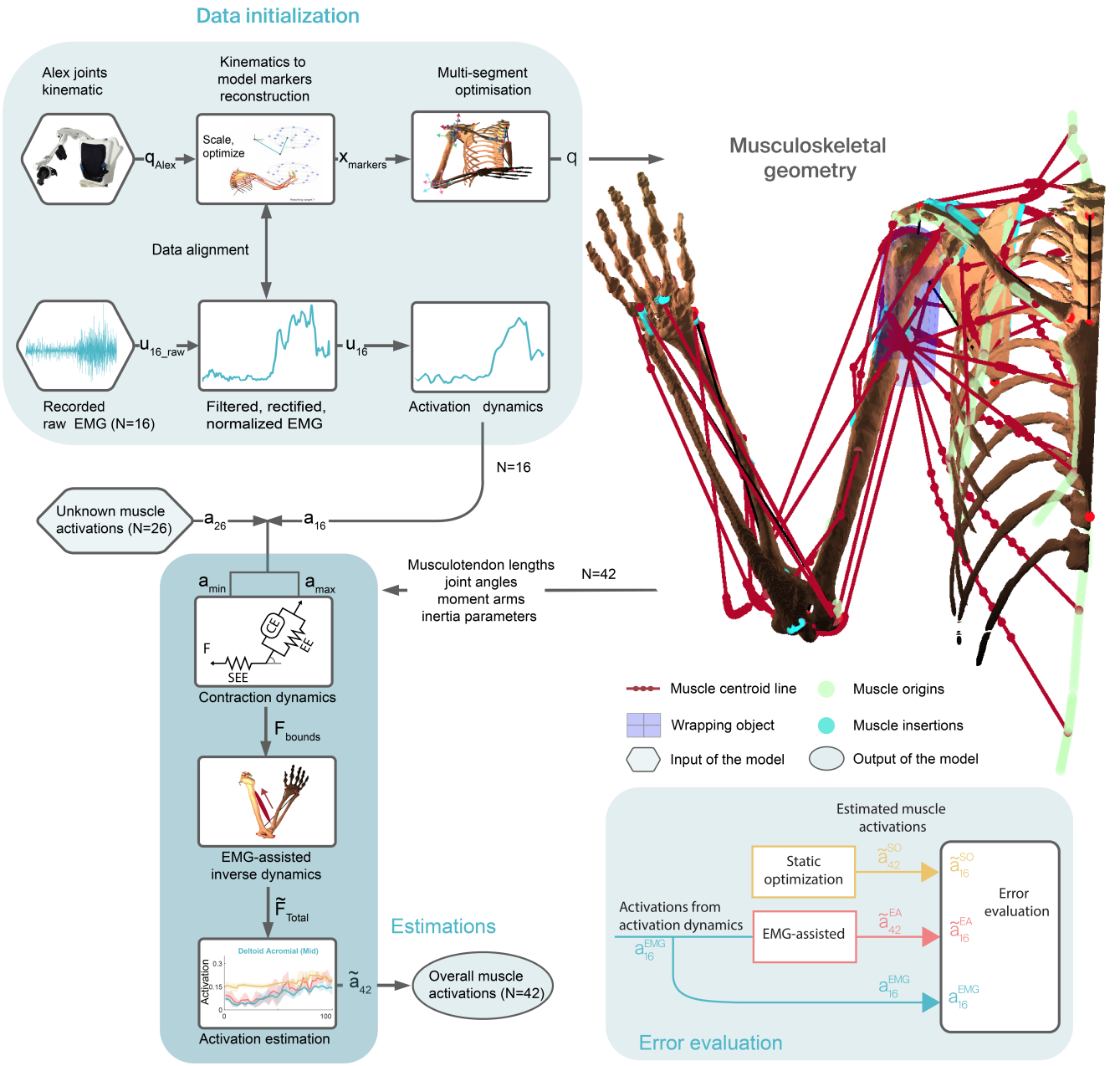


Figure 1. General overview of the presented estimation model. During data initialization, the recorded kinematics q_{ALEx} and EMG signals u_{raw} from N=16 muscles are used to compute the joint angles q and muscle activations a_{16}. These are in turn inputs to the numerical model and the estimation framework to obtain the overall muscle activations $\tilde{a}_{42}$, where the tilde denotes an estimated value. To quantify the errors from the model (error evaluation, *i.e.* coefficient of determination and RMSE), the estimated 16 muscles activations $\tilde{a}^{E}_{16}A$ are compared to the ones directly obtained from EMG measurement $a^{E}_{16}MG$ and to a static-optimization using no EMG data $\tilde{a}^{S}_{16}O$. In the musculoskeletal model representation, the red strings represent the 42 muscles originally considered, from their origin (light green) to their insertion (cyan) with one wrapping object represented around the glenohumeral joint as an example. The modelled muscles are: subclavius, serratus anterior upper/middle/lower, trapezius C1-C6/C7/T1/T2-T7, elevator scapulae, rhomboid minor/major T1-T2/major T3-T4, pectoralis minor/major clavicular/major sternal/major ribs, latissimus dorsi thoracic/lumbar/iliac, deltoid clavicular/acromial/scapular, supraspinatus, infraspinatus, subscapularis, teres minor/major, coracobrachialis, triceps brachii long/medial/lateral, biceps brachii short/long, brachialis, brachioradialis, supinator, pronator Teres, flexor carpi radialis/ulnaris, and extensor carpi radialis long/radialis brevis/ulnaris.

did not provide any assistance to the subject while performing the tasks and only supported its weight[29]. During the robotic assessment, the EMG signals of 16 muscles of the upper-limb were recorded: deltoid clavicular/acromial/scapular, trapezius C1/C7, pectoralis major clavicular, latissimus dorsi thoracic, infraspinatus, triceps brachii long/lateral, biceps brachii short/long, brachialis, brachioradialis and pronator teres. The EMGs were recorded at 1.5kHz using AgCl Kendall surface EMG electrodes with a Desktop DTS wireless system (Noraxon, USA). The electrode placement followed the European recommendations for surface electromyography for non-invasive assessment of muscles (SENIAM) and anatomical guidelines[31,32]. Regarding the kinematic data, the robotic joint angles (shoulder abduction, rotation, flexion elbow flexion, prono-supination), as well as the end-effector position and velocities, were acquired at a 1kHz. The kinematic data acquisition was synchronized with the EMG signals recordings via trigger signals sent from ALEx RS when the movement started and stopped.

Data initialization

Matching ALEx joints kinematics to the model markers

Given that the acquired kinematics from ALEx RS did not correspond to the musculoskeletal model markers requirements[21,22], assumptions had to be made so that the model matched the subject's movement. First, the position of the glenohumeral joint was considered fixed in the workspace: the arm and thorax were attached to the robotic exoskeleton leaving a reduced possibility for the rotations and translations of the shoulder complex. Moreover, the design of the shoulder rotation mechanism[23] uses a remote center of rotation, which corresponds to the glenohumeral joint center, allowing the exoskeleton kinematics to be similar to the one of the subjects. Second, the position of the subject's arm with respect to the robotic exoskeleton was also fixed, thanks to the rigid anchoring that does not allow for translation during the motion. Although we know that these assumptions reduce the accuracy of the activation predictions for a single trial, they would still ensure that all the possible movements performed by each subject during the different movement directions and repetitions could be similarly numerically reconstructed.

The robot kinematics (joint angles and end-effector position), together with robot arm lengths, allowed to obtain the skeletal positions with respect to the initial acromion position. In particular, we reconstructed a wireframe including estimated positions of the glenohumeral (GH) point, set as fixed, anatomical landmarks radial styloid (RS), ulnar styloid (US) and lateral and medial elbow (EL, EM). The GH joint is modelled as a ball and socket joint corresponding to three shoulder rotations: abduction, rotation, and flexion around the robotic joint axes 1, 2, and 3 corresponding to the X, Y, and Z axes respectively, following the Cardan angle convention and the length of the robotic upper arm. The robotic elbow could be obtained, derived from it and based on the generic model distance between the EM and EL, the latter position could be estimated. With the elbow flexion/extension as a pin around the joint number 4 (Z-Axis) and the length of the forearm, the wrist position was obtained (corresponding to the joint number 5, Y-axis) and the US point, assumed to be close to the rotational wrist axis, could be derived. The pronation/supination of the elbow was also considered as a pin joint and together with its rotation and the wrist length in the generic model, the position of the RS point could be obtained from the US point.

The robotic exoskeleton was reconstructed numerically as a verification tool for the overall point and segment positions obtained with respect to the model skeletal ones (Fig. 2). Moreover, the resulting markers' positions and their corresponding skeletal representations were linearly scaled so that the recordings would match the skeletal numerical model. As the result of the markers position estimations and the hypothesis of a fixed GH point, the remaining needed markers of the numerical model could be derived from both the original skeletal positions and the fixed distances between the different remaining markers of the model. This procedure works only with the hypothesis of a fixed GH point; the evaluations of the internal dynamics in the shoulder girdle limits the model to a general overview, as the scapula dynamic is underestimated.

Once all the bony landmarks were defined and the kinematics of the robot matched the one of the model, the joint angles of the numerical model could be computed with the multi-segment optimization as previously reported[21,22]. These optimization algorithms use the minimization of the distance between measured markers and their associated bony landmark. The output of the kinematic data initialization was the collection of musculoskeletal joints angles, velocities, and accelerations.

EMG signals preprocessing and activation dynamics

The EMG signals recorded at 1.5kHz were processed offline with MATLAB, following standard literature[29,30]. EMG signals were detrended, band-pass filtered first at 50-500Hz with a 7th order Butterworth filter, rectified, then low-pass filtered with a 7th order Butterworth filter with a 10Hz cut-off frequency to obtain the envelope. To ensure a wide amplitude of signals fed to the musculotendon contraction dynamics of the model, the envelope was first normalized by the median computed across all the executed movements, favoring expected signals and avoiding high-amplitude spikes caused by noises. For the Hill-Type contraction dynamics described later, the input signal activation should be between 0 and 1 which is then ensured with the normalization of the EMG by the similarly filtered maximum voluntary contraction recorded for each muscle[29,33]. For this study and for computational complexity cost reasons, the kinematic and EMG recorded were interpolated to a timestep of 0.01 second. Adapted from[33] and presented in previous work on this model[21], the activation dynamics associates the filtered EMG signal, or neural excitation u, to the muscle activation. In other words, it corresponds, if activated, to the increase of calcium concentration by the sarcoplasmic reticulum in the muscle fiber, or if deactivated, its removal. The activation dynamics are

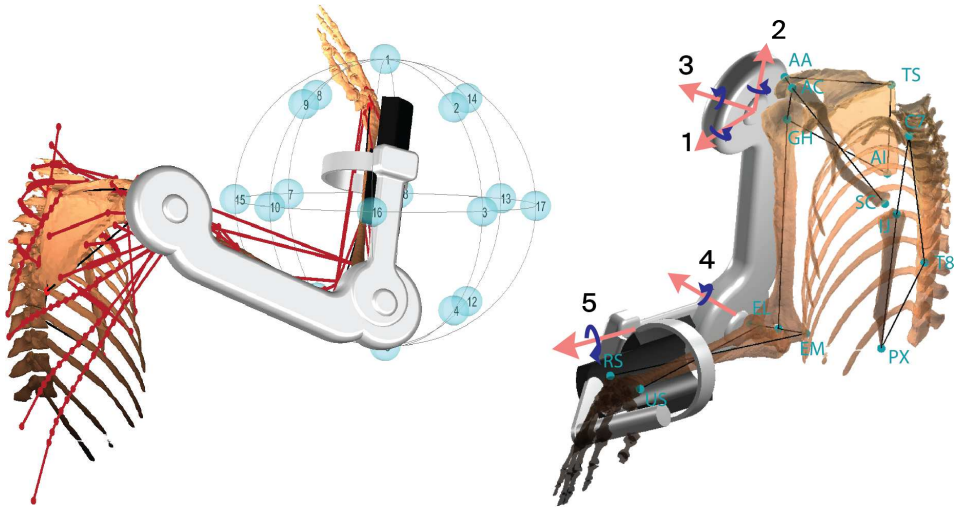


Figure 2. Illustrations representing ALEx RS exoskeleton and musculoskeletal model performing a task in the virtual environment. The blue spheres in the left panel are the different available targets, the red lines are the muscle represented in the model. The different skeletal markers used in the model, right panel, are shown in blue. The numbered robotic joint axes and their rotations are shown in pink.

described in Eq. 1 with the activation a, the excitation u (both between 0 and 1) and τ_{act}, τ_{deact}, the activation and deactivation time constants.

$$\dot{a} = \frac{u-a}{\tau}, \quad \begin{cases} \dfrac{\tau_{act}}{0.5+1.5a} & u < a, \text{activation} \\ \dfrac{\tau_{deact}}{0.5+1.5a} & u \geqslant a, \text{deactivation} \end{cases} \tag{1}$$

Activation estimation

Here we show an estimation of muscle activations in the absence of recorded EMG. The proposed technique is based on an inverse dynamics solver to determine the forces required to produce the torques at each joint, which could account for the observed movement of the subject. Given the over-actuated musculoskeletal model, (42 muscle bundles for 11 DOF), the estimation of the forces involves the optimization of net joint moments among muscles via load-sharing schemes (SLS)[34,35]. However, antagonistic muscles are counterproductive in the net joint moment and SLS overlooks muscle co-contraction[7,14–16,18,36]. The presented musculoskeletal model solves the indeterminacy problem by minimizing the sum of squared muscle stress and constraining the forces within physiological limits[34] which are derived from the available EMG recording[16] and musculotendon parameters (i.e max isometric forces, minimum tendon force from the musculotendon actuation). The overall activation estimation step will be presented in the following order: first, the contraction dynamics and the estimation of the musculotendon parameters will be presented. The EMG-assisted inverse dynamics will be then explained with the resulting activations computed.

Contraction dynamics

The musculotendon actuation links the muscle state and its activation to a produced force. A hill-type model is used to describe the contraction dynamics[16,33,37,38] and consists of three different elements replicating the force production function of the musculotendon system: the contractile element (CE), the passive elastic element (PEE), and serial elastic element (SEE). Each of these elements produces a force defined with normalized functions depending on the muscle intrinsic parameters as presented in the original model[21]. This results in the ordinary differential equation for each muscle Eq. 2.

$$F_{CE}(l_{CE}, v_{CE}, a) + F_{PEE}(l_{CE}) = F_{SEE}(l_{CE}, l_{MTC}) \tag{2}$$

The inputs of the model are the musculotendon length l_{MTC} and activations a together with the fiber lengths and velocities, respectively l_{CE} and v_{CE}. Moreover, although the musculoskeletal model has an elastic tendon model feature incorporated, to allow a wider range of possible solutions for the overall muscle forces, a rigid tendon model was used. This would limit the errors and avoid not feasible solutions from the inverse dynamics as well as considerably reduce the simulation time, which is a

critical element for a model developed for an extended study of muscle activations across a big dataset. In a rigid tendon model, the muscle fiber length l_{CE} can be directly obtained from the musculotendon length l_{MTC} without an extra optimization. In addition to the computational cost reduction, an elastic tendon model increases the importance of perfectly scaled physiological musculotendon parameters. However, obtaining those would require the knowledge of the subject's physiological muscle and skeletal lengths or we would need to use, as it is currently done in recent literature, an optimization on those parameters from the output desired torques to fit a trained model[12–14,39]. However, given the long-term purpose of the proposed model to analyze changes in training and improvement after rehabilitation, such an optimization would lead to different outcomes if performed at the early stages or at the end of rehabilitation. More importantly, if performed for each of the assessments such an optimization would lead to different muscle parameters values which would not be possible in real life: the muscle fiber and tendon lengths should not be altered before and after rehabilitation for instance. With Eq. 2 and the rigid tendon modelization, the muscle fiber and its contraction velocity could be directly obtained from the musculotendon lengths and velocities, leading to the following equation for the musculotendon force produced:

$$\boldsymbol{f}_{MT}(a,\tilde{l}^M,\tilde{v}^M) = F_0[af^L(\tilde{l}^M)f^V(\tilde{v}^M)+f^P(\tilde{l}^M)]\cos\alpha \tag{3}$$

With $\tilde{l}^M$ and $\tilde{v}^M$ the fiber length and velocity normalized by the muscle optimal fiber length and maximal contraction velocity respectively, F_0 the muscle maximal force at optimal fiber length, and f^L, f^V and f^P the normalized function derived from the force-length, force velocity and passive forces relationships obtained by curve fitting of previous experimental data[16], available in the supplementary materials.

The contraction dynamics model gives the boundary values for the physiological muscle forces produced in the EMG-assisted inverse dynamics. The minimal force F_{MT}^{low} produced should correspond to the maximum between the force obtained with an activation of 0 and the passive elastic one, and its maximal F_{MT}^{up} should correspond to the solution with a fully activated contractile element. The lower activation was set to 1e-5 for later mathematical purposes and to avoid numerical discrepancies. In previous work[16], the muscle forces without using EMG data have boundaries between 0 and a Fick constant k= $33Nm^{-2}$ times the muscle physiological cross-sectional area (PCSA) which does not take into consideration the muscular dynamics over the movement. In this study, however, they are defined based on the contraction dynamics, which significantly affects their values, particularly due to the previously neglected passive elastic component.

The musculotendon parameters such as the maximum isometric force, optimal fiber length, pennation angle, maximum velocity contraction, and tendon slack length are adapted from the musculoskeletal model of Garner and Pandy[40] as the muscular architecture from our original model is the same. Using the reported methods[33,41,42], it was then possible to obtain those parameters adapted for the range of muscle lengths across all the possible joint rotations in our model so that a solution always exists for the musculotendon actuation. More precisely, we aimed at keeping the same operating range predicted for each muscle as the ones presented in another study[41], this means that we kept the same ratios of the normalized minimum and maximum length over the muscle optimal fiber length. This ensures a solution for all possible motions of the musculoskeletal model and thus, with the given scaling of the recordings to the generic model, a possible comparison between the different subjects is possible.

EMG-assisted inverse dynamics

As mentioned before, the musculoskeletal system is defined with redundant muscles acting on the same degrees of freedom, this means that in the equations of motions, there are more unknowns than number of degrees of freedom. To estimate the muscle forces of the over-actuated musculoskeletal model, optimizations among muscles via static-optimization (SO) are usually performed. However, given the presence of antagonistic muscles, co-contraction among them can be overlooked in static-optimization as their net joint moment would compensate each other. The aim for the model would be to simulate the central nervous system criteria on muscle recruitment when performing a movement.

The presented musculoskeletal model solves the indeterminacy of the load-sharing problem thanks to the following optimizations. The main optimization criterion is the reduction of the muscle fatigue that is represented as the reduction of the overall muscle stresses squared $\sum\sigma^2$[40,43] as it was shown to give the best estimation of the muscle activations[44].

The cost function to minimize thus is:

$$C_s = \sum\sigma^2 = \boldsymbol{f}^T P\boldsymbol{f} \tag{4}$$

With σ the muscle stresses, P, the diagonal matrix of inverse squared PCSA, and $\boldsymbol{f}$, the muscle forces magnitudes.

However, the model needs to account for constraints such as the maximal and minimal force that the musculotendon can produce given its actual state. These boundaries values are obtained via the contraction dynamics as mentioned before (Eq. 2). For the 16 muscles with recorded EMG, their derived activations a_{16rec} were obtained via the activation dynamics[21,33] as presented in the supplementary material. Because of inter-variability of the EMG recordings[45], the preprocessing steps and

the multiple treatments of the EMG signals and their normalization, the estimated activations from those signals $\tilde{a}_{16}$ were not strictly set. Thus an error was made possible but had to be minimized, with a cost function corresponding to:

$$\hat{E}_{a_{EMG}} = \sum |\boldsymbol{a}_{16rec} - \tilde{\boldsymbol{a}}_{16}| \tag{5}$$

This choice was made to maximize the feasible solutions from the inverse dynamics while keeping a constant process for intra-subject, muscle and movement comparisons. This can not only compensate for possible inaccuracies from EMG signals but also adapt the solution to the overall feasible activities defined by the movement. It ensures a limited deviation from the activation obtained from the activation dynamics for the recorded muscles Given the complexity of the movement recorded with the unconstrained 3D reaching task, some flexibility was permitted regarding the dynamics of the model. Indeed, a feasible solution within the force bounds satisfying the dynamics of the movement could not always be found. This flexibility was also introduced since there might be some inaccuracies in the computed joint angles around the scapulothoracic and acromioclavicular joints given the limited set of obtained positions from the exoskeleton.

The cost corresponding function becomes:

$$\hat{E}_J = \sum |W_m \boldsymbol{f} - \boldsymbol{J}_{torque}| \tag{6}$$

With $\boldsymbol{J}_{torque}$ the torque at the joint satisfying the equations of movement, W_m, the moment arm matrix, corresponding to the cross product between the lever arms and the force direction vectors. The Moment arm matrix, is of size (11x44) defined in the inertial frame and its product with the vector of the 44 force magnitudes $\boldsymbol{f}$ gives the resulting torque from the forces to the joint. Indeed, although there are 42 muscles studied, these elements are augmented with the scapulothoracic contact constraints ($[W_{m_{TS}}\, W_{m_{AI}}][\lambda_{TS}\, \lambda_{AI}]^T$) ensuring the gliding of the scapula over the ribcage as defined for the model in previous work[16,21,22].

Finally the overall cost function to minimize with the estimated activations becomes:

$$A_{cf} = \hat{E}_J + \alpha \hat{E}_{a_{EMG}} + \beta C_s \tag{7}$$

$$= \sum |W_m \boldsymbol{f} - \boldsymbol{J}_{torque}| + \alpha \sum |\boldsymbol{a}_{rec} - \tilde{\boldsymbol{a}}| + \beta \boldsymbol{f}^T P \boldsymbol{f} \tag{8}$$

The α and β are positive weighting coefficients obtained experimentally ensuring similar order weights for $\hat{E}_J \hat{E}_{a_{EMG}}$ The activations minimizing the cost function are obtained with the *fmincon* algorithm in MATLAB. Of note, the parameters α and β were kept constant for all the simulations as we did not want to introduce a supplementary source of differences between the subjects and the different repetitions. For a faster implementation, the forces from contraction dynamics were precomputed at each time step to be functions of the activations only, greatly reducing the computational cost in the *fmincon* algorithm.

Model results validation

In general, three main methodologies were compared to evaluate the model's outputs, see Fig. 1. The first one – *static-optimization* – used no EMG data in the computation of the muscle boundaries for the inverse dynamics, which are therefore obtained with a 0 or 1 muscle activation in the musculotendon actuation (low and upper bound respectively). The second one – *EMG-assisted* – used for the recorded muscles their activations obtained from the activation dynamics to compute the musculotendon actuation. For muscle without available EMG, the boundaries are obtained with a 0 or 1 muscle activation (16 muscle activations used to estimate the overall 42 muscle bundles activations). Finally, for the third one – *from-activation-dynamics* – the activations were directly obtained via the activation dynamics from the recordings and represent the ground truth. This last method is however only available for the 16 muscles with recorded EMG, thus in the following results when those 16 muscles are evaluated, only their corresponding 16 estimations from the static-optimization and EMG-assisted methods were used.

A leave one-out signal comparison (including 15 out of the 16 EMG data) was performed to further validate the estimation process and to have an overview of its robustness. To validate the signal outputs of the model and to have a quantified metric of the similarity between estimated signals and the ones considered as ground truth, a correlation analysis was also performed. For sake of readability, only a selection of the most informative figures representing the results of the simulation and of the comparison is displayed; for a full set of simulations and results, please refer to the supplementary material.

Results

Raw comparison of activations for a specific target

Fig. 3 shows for a specific reaching target (13), the raw comparison between the means across the repetitions of activations obtained via the static-optimization, the EMG-assisted method, and the activations directly derived from the recorded EMG. The duration of each repetition has been normalized so that the activations are displayed as a function of the percentage of

movement duration. Ideally, we would expect the static optimization model to be different from the EMG-assisted method as a demonstration that EMG data are needed to build a good estimation, to show the central nervous system strategy to perform a movement by reducing the overall stress and activations given movement specificity. The first element that can be observed is that the levels of activations over time for all muscles are similar and consistent between all estimated activations; it is important to note also that there is more variation in the muscles activation estimations from the EMG-assisted method than from the static-optimization for non-recorded muscles suggesting a better representation of co-contractions and dynamics within a movement.



Figure 3. Means of the estimated muscles activations across the 5 repetitions for all 5 subjects while reaching target 13, with respect to the percentage of movement duration. The blue lines correspond to the activation obtained directly via the activation dynamics for muscles with available recorded EMG (their name and frame in blue). The red lines correspond to the ones obtained from the presented method using the available EMG data. The yellow lines are the activations obtained from the method without using the recorded EMG, setting the boundaries in the optimization only using the musculotendon actuation with an activation of 0 and 1 for lower bound and upper bound respectively. The thick lines represent the means of those activations across the 5 repetitions. The shaded areas represent their standard deviations.

Error evaluation in joint moments and activations estimations

To further illustrate the overall consistency of the estimations, the mean value of the coefficient of determination R2 between the static-optimization method and the activations from the EMG-assisted model ones over the whole set of activations obtained is 0.27 ± 0.28 which shows a weak with high variability, statistically significant correlation ($p<0.01$ for the 5 subjects, 6 targets, 5 repetitions each target, and the 42 muscles spanning the shoulder, humerus, ulna, and radius). Table 1 shows per target reached, the mean values of the root mean squared error (RMSE) and the R2 between the estimated activations and the recorded ones (N=16) with respect to the method used. The overall RMSE for the EMG-estimated method is as expected lower than the one from SO with an overall mean of 0.03 ± 0.01 against 0.22 ± 0.04 ($p<0.01$). The same trend is observable with the coefficient of determination R2 reported with a weak, but significant correlation with the SO (mean of 0.21 ± 0.2 and p always <0.01). Its large standard deviation indicates that the correlation coefficients have high variabilities, suggesting that some data points are well correlated while others no. With the EMG-assisted method we can observe strongly correlated signals with an overall mean of 0.91 ± 0.21 and median value of 0.99. To be noticed, the results while reaching target 5 show a

better fit in the static optimization method for the RMSE but a lower one for the R2 with respect to the other targets reached. It corresponds to the reaching downwards and this could be explained by the reduced muscle forces needed to perform the movement due to the exoskeleton's resistive action and its gravity compensation for transparency, as well as the lower activation levels of the static-optimization method. In other words, fewer differences in low activation values but more differences in their shapes, which especially shows the importance of the introduction of EMG data in the estimations. The presented values are also reported with respect to each of the muscles activations in the supplementary materials. Additionally the error from the recomputed moment of force at the joints and the one satisfying the equations of movement corresponding to the cost function minimization of eq. 6 is shown. Here, values are represented as a percentage of their maximum measured joint moment from the equations of movement for each of the targets and repetition to have a clear and comparable understanding. These for both methods were kept low with a mean of $3.39\% \pm 2.13$ for the EMG-assisted and $2.33\% \pm 1.56$. We can notice that this value is lower for the SO ($p<0.01$) as expected since there are fewer constraints on each of the muscles' forces from the contraction dynamics. Figures showing the estimated moments with respect to the ones from equations of movement can be found in the supplementary materials.

Target reached	1	3	5	7	10	13	total
NRMSE Moment EMG-assisted	2.70% ± 1.25	2.30% ± 0.06	4.01% ± 3.05	2.95% ± 1.05	4.77% ± 2.86	3.58% ± 1.82	3.39% ± 2.13
NRMSE Moment static-optimization	1.64% ± 0.42	1.89% ± 0.45	2.77% ± 2.21	1.97% ± 0.52	3.94% ± 2.31	1.79% ± 0.42	2.33% ± 1.56
RMSE activations EMG-assisted	0.04 ± 0.01	0.02 ± 0.00	0.02 ± 0.00	0.04 ± 0.01	0.03 ± 0.01	0.03 ± 0.01	0.03 ± 0.01
RMSE activations static-optimization	0.25 ± 0.04	0.20 ± 0.03	0.15 ± 0.02	0.20 ± 0.03	0.28 ± 0.07	0.28 ± 0.04	0.22 ± 0.04
R2 EMG-assisted	0.92 ± 0.05	0.95 ± 0.04	0.92 ± 0.04	0.87 ± 0.10	0.89 ± 0.05	0.86 ± 0.06	0.91 ± 0.05
R2 static-optimization	0.25 ± 0.24	0.16 ± 0.19	0.11 ± 0.15	0.19 ± 0.19	0.20 ± 0.20	0.32 ± 0.25	0.21 ± 0.22

Table 1. Table representing for each of the 6 targets reached the overall errors from estimations and recorded data. First, the error from the computed moments and the ones needed to satisfy the equations of movement, in both the static-optimization and EMG-assisted cases. Values are normalized by the maximum moment from the equations of movement for each of the target and repetition. We can notice that this value is lower for the SO ($p<0.001$). Then the RMS and coefficient of determination R2 between the activations resulting from the recorded EMG and their estimations via the EMG-assisted and SO methods are reported for each of the targets, As expected, the EMG_assisted method presents a better fit ($p<0.001$ in both cases). We can see the low means of R2 in static-optimization showing a weak correlation, the large standard deviation indicates that the correlation coefficients have high variability, which suggests that the estimated signals may be somewhat correlated with the recorded signals for some data points, but not for others. To be noticed, the Targets 5 shows a reduced fit in both methods with respect to the other targets reached.

Mean activation for all targets

Another way to validate the obtained activation signals is to compare their levels of activation for specific movements. Fig. 4(a) shows these different levels of activations for two muscles with respect to the reached target. From the figure, one can observe that on the one hand, there is a good estimated activation level for some muscles in the static-optimization. On the other hand, this shows its limits, overlooking co-contractions and therefore underestimating values of activations for most muscles. Indeed, the activation levels obtained from the static-optimization were almost always below the ones of the EMG-assisted or the activation dynamics. In any case, the EMG-assisted method provides consistent results with respect to the activations obtained directly from the activation dynamics, and the difference among those values is explained by the fact that the EMG-assisted estimations take the geometrical model state and the contraction dynamics during the motion, and therefore depends on the muscle parameters in addition to the needs from the equations of movement. Regarding the values of the estimated activations, it is clear that there is a good correlation between the expected activation levels and the obtained ones (mean $R2 = 0.89 \pm 0.20$ and median $R2 = 0.99$) as presented in Table 1. In the presented Fig. 4(a), expected higher activation levels were met for brachialis for reaching targets 10 (involving an elbow flexion) for all cases and a lower on for reaching target 5 (elbow extension). It was however less evident for the biceps short which was more used for the upper sagittal plane targets (1,10 and 13). For the static optimization, the biceps short activations were underestimated and not physiologically expected. It seems that the combination of all the synergistic muscle forces produced still induced a torque leading to the correct movement.

Leave-one-out cross validation

Fig. 4(b) represents a spider map showing for each recorded muscle that was removed successively in the leave-one-out cross-validation, the root mean square errors between the activation directly obtained from the activation dynamics and the method using all muscle EMG signals (N=16 in the EMG-assisted method), using all EMG available excluding the specific muscle shown (N=15 in the EMG-assisted method) and without any. The mean RMSE for the EMG-assisted method was

0.03 ± 0.01, for the leave-one-out of 0.17 ± 0.02 and for the static-optimization 0.22 ± 0.08, This shows a slight improvement using the EMG-assisted method for the activation estimations for non-recorded muscles (p=0.03), but most importantly, it seems to ensure a physiological sharing of the load, and to give more information about the muscle's behavior providing a better insight of the actual co-contractions with lower variations in specific estimated activations within same movements, being particularly noticeable with the brachialis with the biceps and the deltoids.

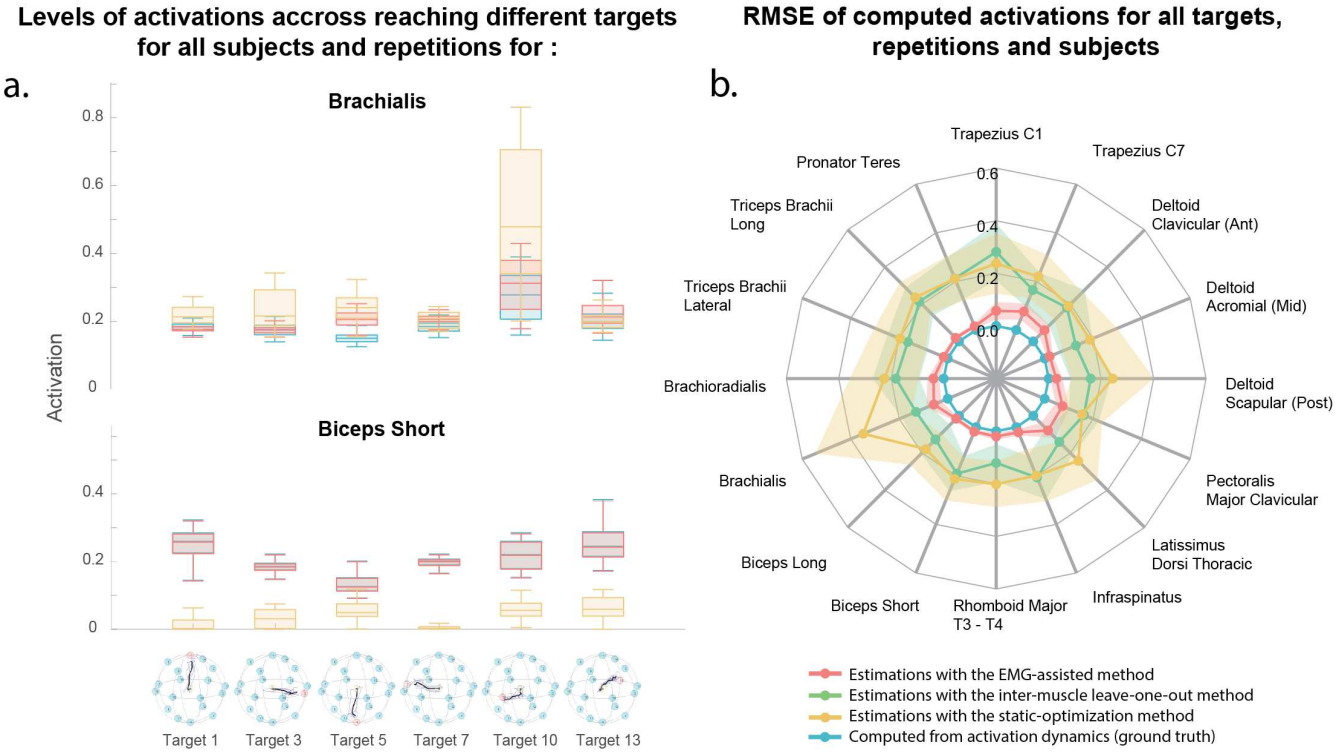


Figure 4. **(a)** Comparisons between the ranges of activations directly obtained from activation dynamics (blue boxplot), the estimated ones with the model including the recorded EMG (red boxplot) and the one from the model without using the recorded EMG (yellow boxplot) with respect to the reached target. The first comparison is for the brachialis and below for the biceps (Short head). The different reached targets are represented, in the bottom, as red spheres and in yellow is shown the starting position, corresponding to the center of the reachable globe. The black line is the means trajectory of the end effector and the thinner blue lines are the 5 repetitions recorded. **(b)** Spider map of the root mean square errors of the computed estimation of the muscle activation through all targets reached and repetitions of subject 2. The figure shows for each muscle with recorded EMG the errors between the activation directly obtained from the activation dynamics and, in red, the method using all the available muscle EMG signals in the EMG-assisted model (16 muscles out of 16). In green, using all EMG available excluding the specific muscle in the EMG-assisted model(15 muscles used out of 16), and without any EMG (in yellow) corresponding to the static-optimization.

Discussion

This work aimed at developing a tool for an overall muscle activation estimation using a limited amount of recorded EMG data. Knowing the inter-subject muscle parameter differences as well as the variability in EMG signal recordings from the same motion, the goal is not to provide an exact estimation of each specific muscle activation, but to infer the scheme of overall muscles activation during a specific task, with the overall values highlighted in Fig. 3 and Fig. 4. We first accurately reproduced with a generic numerical model the trajectory performed by the subjects with the ALEx RS exoskeleton. Then we showed that the EMG-assisted estimation process allowed us to have a clearer insight into the overall activation patterns. This was mostly done by inducing restrictions using recorded EMG signals to the possible forces produced in the contraction dynamics of the model, which resulted in a finer range of solutions to the load sharing problem as it was highlighted also in another EMG-assisted model[19]. As noticeable in Fig. 3, the outcome is a thorough estimation of possible activations when using the EMG-assisted method. Despite the static-optimization model computed activations, with no recorded EMG used in the estimation process, has a similar root mean square error with a mean RMSE of 0.22 ± 0.08 (Fig. 4(b)), the EMG-assisted

model, illustrated with the leave-one-out validation, is still more accurate in providing a general overview about the typical activations with a mean RMSE of 0.17 ± 0.02 for the remaining muscles. More importantly, we can observe less variability in the estimations, showing that its consistency is better and that it is less prone to random fluctuations. This further validates the use of recorded signals in our estimation process. Although the improvement of activation signal estimates using EMG-assisted methods has been demonstrated previously, to the authors' knowledge this has never been performed outside of the OpenSim environment and with a complex shoulder model including 11 DOF and 42 distinct muscles. Moreover, the tasks and movements presented here do not constrain the range of motion and the plane of action, as it was done previously while studying upper limb activations[46] or to a 3 DOF study[14]

The ground truth used in the comparisons is the activation dynamics from recorded EMG[19]. Even though the experimental muscle activations are used in the EMG-assisted model, giving this method an advantage over the static-optimization one in the evaluations, the result of the leave-one-out cross-validation (Fig. 4(b)) indicates that the EMG-assisted model still performs better in estimating muscle activation without any EMG-related information.

While we could obtain estimated muscle activations with reasonable values for all muscles during the reaching task, it is important to emphasize that the contraction dynamics equations for the numerical estimates of forces from excitations are derived from a rigid tendon model. One could argue that with a compliant tendon model for the contraction dynamics we would obtain a more accurate activation. Despite true, this would lead to a supplementary optimization process to obtain at each iteration the muscles' fiber lengths and their tendon lengths[21], not something that can be easily assessed experimentally or without a computational costly optimization[47]. Another possibility would be to use a different tendon model depending on the ratio of the tendon slack length to the optimal fiber length as suggested in another study[48], but for a validation process, we wanted to be consistent within the muscle model used. Finally, the resulting nRMSE between the torques needed for the GH, EL and PS joints in the equations of movement and the ones resulting from both optimisation process were kept low (with means of 3.4% and 2.3% for the EMG-assisted and static-optimization respectively). Even though we could reconstruct the motion of the claviculae and scapula, these were constrained by our initial assumptions, and thus potential motions would be overlooked, reducing the activations estimation relevance. On a general note, the EMG-assisted model predicted more accurate activations for the muscles involved in the movement, with greater co-contractions. Apart from the mean of comparison, using the experimental muscle activations that were used by the EMG-assisted model, this could be explained as the SO method is solely based on the minimization of the inverse squared of muscles' physiological cross-section areas criteria, disregarding possible unique activations patterns between subjects[49]. With a full scaling of all musculotendon parameters and inertia based on subjects' data, which is permitted by the model[21] we could expect even better-tracked activations.

In this study, we gathered motion data from healthy individuals and the resulting model is a promising first step towards a general understanding of the CNS strategy to control the upper-limb. In a future application, we will apply the same methodology to trajectory data from individuals with physical disability (e.g. Parkinson, stroke). Being able to understand patterns of activation, maybe even compare pre- vs post-intervention data in a rehabilitation scenario could be of great interest for the community and open up new possibilities in designing optimized therapy for the patients[50] or lower dimensional controllers for the upper-limb, as it was done with reduced models for the frog hindlimb[51] or human lower limbs[52]. The knowledge of a generalized activations estimation could also be used to further improve motor intention decoding for prosthetic-end users[53] as well as improve personalized prosthesis myoelectric control[6]. In the future, this methodology and model could also potentially be an open tool for everyone to use.

Conclusion

This work presents the development and validation of a musculo-skeletal model usable to increase our basic understanding of upper-limb motor control. It presents a protocol with ALEx RS exoskeleton to assess the overall muscle activity while performing reaching tasks. We first reconstructed the upper-limb motion, then, with the combined recorded EMG signals while reaching tasks, we showed how our model can estimate the overall muscle activation with an EMG-assisted load sharing optimization method. This protocol and the activations estimation represent a tool to evaluate inter-subject muscle activation behavior, possible interactions and synergies. This was applied to healthy motion data but in the future, we will target neurological diseases patient's motion data. Potentially, this could help with the development of novel neurorehabilitation strategies, lower dimensional controllers for the upper-limb, improve motor intention decoding for prosthetic-end users as well as improve personalized prosthesis myoelectric control.

References

1. Bernstein, N. *The co-ordination and regulation of movements* (Pergamon Press, 1967).
2. Mussa-Ivaldi, F. A. & Bizzi, E. Motor learning through the combination of primitives. *Philos. Transactions Royal Soc. London. Ser. B: Biol. Sci.* **355**, 1755–1769, DOI: 10.1098/RSTB.2000.0733 (2000).

3. Bizzi, E. & Cheung, V. C. The neural origin of muscle synergies. *Front. Comput. Neurosci.* **7**, 1–6, DOI: 10.3389/FNCOM.2013.00051/BIBTEX (2013).

4. D'Avella, A., Saltiel, P. & Bizzi, E. Combinations of muscle synergies in the construction of a natural motor behavior. *Nat. Neurosci. 2003 6:3* **6**, 300–308, DOI: 10.1038/nn1010 (2003).

5. Muraoka, T., Muramatsu, T., Takeshita, D., Kanehisa, H. & Fukunaga, T. Estimation of passive ankle joint moment during standing and walking. *J. applied biomechanics* **21**, 72–84, DOI: 10.1123/JAB.21.1.72 (2005).

6. Sartori, M., Durandau, G., Došen, S. & Farina, D. Robust simultaneous myoelectric control of multiple degrees of freedom in wrist-hand prostheses by real-time neuromusculoskeletal modeling. *J. neural engineering* **15**, DOI: 10.1088/1741-2552/AAE26B (2018).

7. Cholewicki, J., McGill, S. M. & Norman, R. W. Comparison of muscle forces and joint load from an optimization and emg assisted lumbar spine model: towards development of a hybrid approach. *J. biomechanics* **28**, DOI: 10.1016/0021-9290(94)00065-C (1995).

8. Kaufman, K. R., An, K. N., Litchy, W. J. & Chao, E. Y. Physiological prediction of muscle forces–i. theoretical formulation. *Neuroscience* **40**, 781–792, DOI: 10.1016/0306-4522(91)90012-D (1991).

9. Happee, R. Inverse dynamic optimization including muscular dynamics, a new simulation method applied to goal directed movements. *J. Biomech.* **27**, 953–960, DOI: 10.1016/0021-9290(94)90267-4 (1994).

10. Rasmussen, J., Damsgaard, M. & Voigt, M. Muscle recruitment by the min/max criterion – a comparative numerical study. *J. biomechanics* **34**, 409–415, DOI: 10.1016/S0021-9290(00)00191-3 (2001).

11. Tamilselvam, Y. K., Ganguly, J., Patel, R. V. & Jog, M. Musculoskeletal model to predict muscle activity during upper limb movement. *IEEE Access* **9**, 111472–111485, DOI: 10.1109/ACCESS.2021.3103438 (2021).

12. Sartori, M., Farina, D. & Lloyd, D. G. Hybrid neuromusculoskeletal modeling to best track joint moments using a balance between muscle excitations derived from electromyograms and optimization. *J. Biomech.* **47**, 3613–3621, DOI: 10.1016/J.JBIOMECH.2014.10.009 (2014).

13. Lloyd, D. G. & Besier, T. F. An emg-driven musculoskeletal model to estimate muscle forces and knee joint moments in vivo. *J. Biomech.* **36**, 765–776, DOI: 10.1016/S0021-9290(03)00010-1 (2003).

14. Assila, N., Pizzolato, C., Martinez, R., Lloyd, D. G. & Begon, M. Emg-assisted algorithm to account for shoulder muscles co-contraction in overhead manual handling. *Appl. Sci. 2020, Vol. 10, Page 3522* **10**, 3522, DOI: 10.3390/APP10103522 (2020).

15. Gagnon, D., Arjmand, N., Plamondon, A., Shirazi-Adl, A. & Larivière, C. An improved multi-joint emg-assisted optimization approach to estimate joint and muscle forces in a musculoskeletal model of the lumbar spine. *J. Biomech.* **44**, 1521–1529, DOI: 10.1016/J.JBIOMECH.2011.03.002 (2011).

16. Sarshari, E. *et al.* Muscle co-contraction in an upper limb musculoskeletal model: Emg-assisted vs. standard load-sharing. *https://doi.org/10.1080/10255842.2020.1814755* **24**, 137–150, DOI: 10.1080/10255842.2020.1814755 (2020).

17. Kian, A. *et al.* Static optimization underestimates antagonist muscle activity at the glenohumeral joint: A musculoskeletal modeling study. *J. biomechanics* **97**, DOI: 10.1016/J.JBIOMECH.2019.109348 (2019).

18. Nikooyan, A. A. *et al.* An emg-driven musculoskeletal model of the shoulder. *Hum. Mov. Sci.* **31**, 429–447, DOI: 10.1016/J.HUMOV.2011.08.006 (2012).

19. Hoang, H. X., Diamond, L. E., Lloyd, D. G. & Pizzolato, C. A calibrated emg-informed neuromusculoskeletal model can appropriately account for muscle co-contraction in the estimation of hip joint contact forces in people with hip osteoarthritis. *J. Biomech.* **83**, 134–142, DOI: 10.1016/J.JBIOMECH.2018.11.042 (2019).

20. Pizzolato, C. *et al.* Ceinms: A toolbox to investigate the influence of different neural control solutions on the prediction of muscle excitation and joint moments during dynamic motor tasks. *J. Biomech.* **48**, 3929–3936, DOI: 10.1016/J.JBIOMECH.2015.09.021 (2015).

21. Sarshari, E. *A Closed-Loop EMG-Assisted Shoulder Model*. Ph.D. thesis, EPFL (2018). DOI: 10.5075/EPFL-THESIS-8658.

22. Ingram, D. *Musculoskeletal Model of the Human Shoulder for Joint Force Estimation*. Ph.D. thesis, EPFL (2015). DOI: 10.5075/EPFL-THESIS-6497.

23. Lenzo, B., Bergamasco, M. & Salsedo, F. Actuating method and device for human interaction multi-joint mechanisms (WO Patent App, WO2013186701, 2015).

24. Curreli, C., Puccio, F. D., Davico, G., Modenese, L. & Viceconti, M. Using musculoskeletal models to estimate in vivo total knee replacement kinematics and loads: Effect of differences between models. *Front. Bioeng. Biotechnol.* **9**, 611, DOI: 10.3389/FBIOE.2021.703508/BIBTEX (2021).

25. Modenese, L., Ceseracciu, E., Reggiani, M. & Lloyd, D. G. Estimation of musculotendon parameters for scaled and subject specific musculoskeletal models using an optimization technique. *J. Biomech.* **49**, 141–148, DOI: 10.1016/j.jbiomech.2015.11.006 (2016).

26. Monsabert, B. G. D., Edwards, D., Shah, D. & Kedgley, A. Importance of consistent datasets in musculoskeletal modelling: A study of the hand and wrist. *Annals Biomed. Eng.* **46**, 71, DOI: 10.1007/S10439-017-1936-Z (2018).

27. Wu, G. *et al.* Isb recommendation on definitions of joint coordinate systems of various joints for the reporting of human joint motion–part ii: shoulder, elbow, wrist and hand. *J. biomechanics* **38**, 981–992, DOI: 10.1016/J.JBIOMECH.2004.05.042 (2005).

28. Ingram, D., Engelhardt, C., Farron, A., Terrier, A. & Müllhaupt, P. Muscle moment-arms: a key element in muscle-force estimation. *Comput. methods biomechanics biomedical engineering* **18**, 506–513, DOI: 10.1080/10255842.2013.818666 (2015).

29. Pirondini, E. *et al.* Evaluation of the effects of the arm light exoskeleton on movement execution and muscle activities: A pilot study on healthy subjects. *J. NeuroEngineering Rehabil.* **13**, 1–21, DOI: 10.1186/S12984-016-0117-X/FIGURES/7 (2016).

30. Pierella, C. *et al.* A multimodal approach to capture post-stroke temporal dynamics of recovery. *J. neural engineering* **17**, DOI: 10.1088/1741-2552/AB9ADA (2020).

31. Hermens, H. J., Freriks, B., Disselhorst-Klug, C. & Rau, G. Development of recommendations for semg sensors and sensor placement procedures. *J. electromyography kinesiology : official journal Int. Soc. Electrophysiol. Kinesiol.* **10**, 361–374, DOI: 10.1016/S1050-6411(00)00027-4 (2000).

32. Delagi, E. F. & Perotto, A. *Anatomic guide for the electromyographer : the limbs / by Edward F. Delagi, Aldo Perotto ; with a foreword by Arthur S. Abramson ; illustrated by Hugh Thomas.* (Thomas, Springfield, Ill, 1980), 2d ed. edn.

33. Zajac, F. E. Muscle and tendon: properties, models, scaling, and application to biomechanics and motor control. *Critical reviews biomedical engineering* **17**, 359—411 (1989).

34. Terrier, A. *et al.* A musculoskeletal shoulder model based on pseudo-inverse and null-space optimization. *Med. engineering & physics* **32**, 1050–1056, DOI: 10.1016/J.MEDENGPHY.2010.07.006 (2010).

35. Herzog, W. Force-sharing among synergistic muscles. *Exerc. Sport Sci. Rev.* **24**, 173–202, DOI: 10.1249/00003677-199600240-00008 (1996).

36. Collins, J. J. The redundant nature of locomotor optimization laws. *J. biomechanics* **28**, 251–267, DOI: 10.1016/0021-9290(94)00072-C (1995).

37. Hill, A. V. The heat of shortening and the dynamic constants of muscle. *Proc. Royal Soc. B: Biol. Sci.* **126**, 136–195, DOI: 10.1098/RSPB.1938.0050 (1938).

38. Arslan, Y. Z., Karabulut, D., Ortes, F. & Popovic, M. B. Exoskeletons, exomusculatures, exosuits: Dynamic modeling and simulation. *Biomechatronics* 305–331, DOI: 10.1016/B978-0-12-812939-5.00011-2 (2019).

39. Sartori, M., Reggiani, M., Farina, D. & Lloyd, D. G. Emg-driven forward-dynamic estimation of muscle force and joint moment about multiple degrees of freedom in the human lower extremity. *PLOS ONE* **7**, e52618, DOI: 10.1371/JOURNAL.PONE.0052618 (2012).

40. Garner, B. A. & Pandy, M. G. Musculoskeletal model of the upper limb based on the visible human male dataset. *Comput. methods biomechanics biomedical engineering* **4**, 93–126, DOI: 10.1080/10255840008908000 (2001).

41. Garner, B. A. & Pandy, M. G. Estimation of musculotendon properties in the human upper limb. *Annals Biomed. Eng. 2003 31:2* **31**, 207–220, DOI: 10.1114/1.1540105 (2003).

42. Hainisch, R., Gfoehler, M., Zubayer-Ul-Karim, M. & Pandy, M. G. Method for determining musculotendon parameters in subject-specific musculoskeletal models of children developed from mri data. *Multibody Syst. Dyn.* **28**, 143–156, DOI: 10.1007/S11044-011-9289-0/FIGURES/5 (2012).

43. Crowninshield, R. D. & Brand, R. A. A physiologically based criterion of muscle force prediction in locomotion. *J. Biomech.* **14**, 793–801, DOI: 10.1016/0021-9290(81)90035-X (1981).

44. Luis, I., Afschrift, M., Groote, F. D. & Gutierrez-Farewik, E. M. Evaluation of musculoskeletal models, scaling methods, and performance criteria for estimating muscle excitations and fiber lengths across walking speeds. *Front. Bioeng. Biotechnol.* **10**, 1865, DOI: 10.3389/FBIOE.2022.1002731/BIBTEX (2022).

45. Hug, F. Can muscle coordination be precisely studied by surface electromyography? *J. electromyography kinesiology : official journal Int. Soc. Electrophysiol. Kinesiol.* **21**, 1–12, DOI: 10.1016/J.JELEKIN.2010.08.009 (2011).

46. Ashari, M. F., Hanafusa, A. & Mohamaddan, S. Evaluation of upper limb muscle activation using musculoskeletal model with wearable assistive device. *Appl. Bionics Biomech.* **2022**, DOI: 10.1155/2022/8908061 (2022).

47. Millard, M., Uchida, T., Seth, A. & Delp, S. L. Flexing computational muscle: modeling and simulation of musculotendon dynamics. *J. biomechanical engineering* **135**, DOI: 10.1115/1.4023390 (2013).

48. Mousavi, A., Ehsani, H. & Rostami, M. Compliant vs. rigid tendon models: A simulation study on precision, computational efficiency and numerical stability. *2014 21st Iran. Conf. on Biomed. Eng. ICBME 2014* 92–97, DOI: 10.1109/ICBME.2014.7043900 (2011).

49. Buchanan, T. S. & Shreeve, D. A. An evaluation of optimization techniques for the prediction of muscle activation patterns during isometric tasks. *J. Biomech. Eng.* **118**, 565–574, DOI: 10.1115/1.2796044 (1996).

50. Sartori, M., Llyod, D. G. & Farina, D. Neural data-driven musculoskeletal modeling for personalized neurorehabilitation technologies. *IEEE transactions on bio-medical engineering* **63**, 879–893, DOI: 10.1109/TBME.2016.2538296 (2016).

51. Berniker, M., Jarc, A., Bizzi, E. & Tresch, M. C. Simplified and effective motor control based on muscle synergies to exploit musculoskeletal dynamics. *Proc. Natl. Acad. Sci. United States Am.* **106**, 7601–7606, DOI: 10.1073/PNAS.0901512106/SUPPL_FILE/0901512106SI.PDF (2009).

52. Neptune, R. R., Clark, D. J. & Kautz, S. A. Modular control of human walking: a simulation study. *J. biomechanics* **42**, 1282–1287, DOI: 10.1016/J.JBIOMECH.2009.03.009 (2009).

53. Mendez, V., Iberite, F., Shokur, S. & Micera, S. Current solutions and future trends for robotic prosthetic hands. *Annu. Rev. Control. Robotics, Auton. Syst.* **4**, 595–627, DOI: 10.1146/ANNUREV-CONTROL-071020-104336 (2021).

Author contributions statement

T.B conceived the work on the numerical model, its development, the initial draft and the figures. T.P provided critical insights and contributed to the final paper. C.P designed the experimental protocol and conducted the experiments. S.M directed the study. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- ActivationEstimationsSupplementaryMaterial.pdf